

PROBLEM STATEMENT #28 | Smart Cities, Transport & Logistics**Inter-City Freight Load Matching Marketplace****1. Problem Context & Overview**

A B2B freight brokerage exchange where commercial shippers post cargo loads and trucking carriers submit bids, with automated milestone payment release upon proof-of-delivery sign-off.

Target Stakeholders / Actors: *Shipper, Freight Carrier*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall allow verified trucking carriers to submit competitive bids on posted freight shipments within a specified auction timeframe.

Sample Acceptance Criteria: Pass: Carrier bid recorded and shipper notified; Fail: Unverified carrier submits bid without valid permits.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: Milestone payments held in escrow shall disburse automatically within 60 seconds of verified electronic proof-of-delivery submission.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.